# Deep Learning-Based Multi-Student Behaviour Analysis for Emotionally Intelligent Robot Teacher Systems


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**Abstract—**Automated monitoring of student behaviour in large classroom environments presents significant challenges for human instructors and existing computer vision systems alike. Manual supervision is intractable at scale, while prior deep learning approaches either lack edge deployability, suffer from temporal instability, or fail to combine complementary visual and geometric cues. This paper presents a real-time, edge-deployable multi-student behaviour analysis system designed to serve as a perceptual backbone for emotionally intelligent robot teacher platforms. The proposed system integrates a YOLOv9-based face detector with DeepSORT multi-object tracking to achieve persistent student identity assignment across video frames. Sleeping detection is formulated as a dual-stream classification problem: a MobileNetV2 backbone augmented with a learned EyeAttention spatial module processes 224×224 face crops, while a parallel geometric stream encodes a 7-dimensional Eye Aspect Ratio (EAR) feature vector derived via MediaPipe FaceMesh. Features from both streams are fused and passed through a joint classifier, yielding a frame-level sleeping detection accuracy of 96.98%, recall of 100%, and AUC of 0.9863 — with zero false negatives. Talking detection employs a MouthAttention-enhanced MobileNetV2 model operating on cropped mouth regions, achieving a frame-level F1-score of 0.82. A behaviour-hierarchy temporal smoother resolves per-track label assignments across frames. Person-level evaluation across five test videos (31 subjects, 787 frames) yields 77.42% overall accuracy with a sleeping recall of 100%. The system is deployed on a Raspberry Pi 4 with Flask-based MJPEG streaming at 1 Hz inference.



**Keywords—**student behaviour recognition; deep learning; YOLOv9; DeepSORT; MobileNetV2; Eye Aspect Ratio; attention mechanism; edge deployment; classroom monitoring.


## I. Introduction

The extensive growth of large lecture-hall classrooms and the growing demand for personalised, adaptive teaching have created a convincing need for intelligent systems capable of passively monitoring student engagement, alertness, and participation. Human instructors cannot simultaneously track the behavioural state of dozens of students in real time — particularly under tiered seating, variable lighting, and significant occlusion.

Real-time awareness of the environment is a necessity for adaptive classroom-based responses of robot teachers with emotional intelligence, such as those developed on platforms like NAO or Pepper. If a robot can’t tell if the students are paying attention, sleeping or having side conversations, how can it adjust the tempo, volume or content of its lessons?

Existing deep learning approaches to classroom behaviour recognition present several well-documented limitations. YOLO-based end-to-end detection methods [1], [2] achieve competitive mAP scores but their computational complexity delivers them unsuitable for resource-constrained edge devices. Methods relying exclusively on visual CNN features [3] suffer from sensitivity to pose variation and illumination changes. The few lightweight architectures proposed [4] reduce robustness in crowded settings. Additionally, the research literature largely neglects temporal continuity: frame-wise predictions exhibit high variance, yet most systems lack explicit smoothing mechanisms tied to stable identity tracks.

This work addresses these gaps by proposing a modular, multi-stream behavioral analysis pipeline that is both technically rigorous and practically deployable. The major contributions are:

- A dual-stream sleeping detection architecture (SleepingModelV4) fusing deep visual features from a MobileNetV2 backbone with 7-dimensional EAR geometric features, achieving 96.98% accuracy and 100% recall.
- A novel EyeAttention spatial attention module with an anatomical prior that biases feature pooling toward the eye region, improving robustness to partial occlusion.
- A MouthAttention-enhanced TalkingModel operating on anatomically cropped mouth regions, achieving a frame-level F1-score of 0.82 without audio input.
- Integration of YOLOv9 face detection with DeepSORT tracking, achieving zero identity switches across all test videos.
- A per-track behaviour-hierarchy temporal smoother that eliminates noisy frame-wise label oscillations.
- Successful Raspberry Pi 4 deployment with Flask MJPEG streaming and CSV logging, validated at 1 Hz CPU-only inference.

## II. RELATED WORK

### *A. YOLO-Based Behaviour Detection*

Yang et al. [1] proposed an improved YOLOv7 with spatio-temporal attention and Wise-IoU loss for classroom behaviour detection. Despite strong frame-level performance, sequential predictions do show significant label oscillation without smoothing. Fang et al. [2] extended this line by employing a YOLOv11 detector for multi-class real-time detection at competitive rates, but with a considerable need for GPU resources, which hinders its deployment at the edge. Wang and Fan [5] proposed YOLOv11-ASV with further architectural improvements; however, the model size and hardware requirements are still incompatible with single-board computers.

### *B. Multi-Scale and Temporal Architectures*

Zhang et al. [3] proposed a multi-scale spatio-temporal attention architecture, MSTA-SlowFast, for slow and fast frame streams. Although it explicitly models temporal dynamics, its computational weight makes it completely unfit for edge deployment. Gao and Hang [6] tackled efficiency via architectural pruning of YOLOv8s but offer limited temporal modelling without tracking-aware smoothing.

### *C. CNN and Geometric Feature Methods*

Zhang and Li [4] introduced DSCNN, a depthwise separable CNN for lightweight behaviour recognition, noting reduced robustness in crowded scenarios. Critically, none of the above methods combine CNN visual features with geometric facial landmarks - a complementary modality that provides pose-invariant information about eye and mouth aperture that CNN features alone may not reliably encode under challenging illumination.

### *D. Eye Aspect Ratio and Drowsiness Detection*

The EAR was first introduced by Soukupova and Cech [7] for real-time blink detection and then used for drowsiness monitoring in driver alertness systems [8]. EAR-only methods are brittle in the presence of large head pose or low resolution where landmark detection fails. Several fusion approaches [9] demonstrated that the combination of EAR and visual CNN features results in more robust classifiers than either of the modalities alone. In the present work , we generalise this knowledge to classroom sleeping detection with an explicit spatial attention mechanism .

### *E. Research Gap*

Through systematic review, we identify three understudied challenges: (1) lack of lightweight, edge-deployable systems with competitive accuracy, (2) lack of temporal smoothing linked to stable identity tracks, and (3) limited adoption of CNN-EAR fusion for classroom-specific sleeping detection. The present work directly addresses all three gaps.

## III. PROPOSED METHODOLOGY

The proposed system is a modular pipeline in which video frames are processed through face detection, identity-preserving tracking, dual-stream behaviour classification, temporal smoothing, and deployment output stages.

### *A. System Architecture Overview*

Video frames are taken from a video input or webcam or Raspberry Pi PiCamera2. Each frame is passed to a YOLOv9 face detector which returns bounding boxes in [x1, y1, x2, y2] format. Detections are passed to DeepSORT, which assigns persistent integer track IDs consistent across frames. For each confirmed track a face crop is extracted, padded by 8 px, and resized to 224×224. This crop is processed by SleepingModelV4 and conditionally by TalkingModel. A per-track temporal smoother converts frame-wise probabilities into stable behavioural labels. Annotated output and CSV logs are produced continuously.

### *B. Face Detection: YOLOv9*

Face detection uses a YOLOv9 model fine-tuned on face imagery (YOLOv9face.pt). YOLOv9 proposes Programmable Gradient Information (PGI) and the Generalised Efficient Layer Aggregation Network (GELAN) to address information bottleneck problems of deep networks [11]. The detector was configured with input resolution 640×640, confidence threshold 0.60, and a minimum face size of 48×48 pixels. The pipeline outputs a bounding box [x1, y1, x2, y2] and confidence score for each face.

### *C. Multi-Person Tracking: DeepSORT*

We use DeepSORT [12] for persistent identity assignment, which combines a Kalman-filtered motion state with a deep appearance embedding. For each new detection, the Hungarian algorithm solves the bipartite assignment problem with a weighted sum of the Mahalanobis distance (for motion) and the cosine distance (for appearance). In the evaluation of 31 student tracks and 787 frames, DeepSORT achieved zero identity switches, which is a crucial requirement for the temporal smoother to correctly accumulate each student's behavioural evidence.

### *D. Sleeping Detection: SleepingModelV4*

Sleeping detection is based on a dual-stream architecture where each 224×224 face crop is processed by a visual CNN stream and a geometric EAR stream, and then their features are fused.

CNN Stream Visual. The face crop is input to a pretrained MobileNetV2 backbone [14] that outputs a spatial feature map of shape [B, 1280, 7, 7]. We selected MobileNetV2 due to its good trade-off between accuracy and number of parameters and its inverted residual design that makes it suitable for deployment at the edge. This feature map passes through the EyeAttention module.

Eye Attention Module. Standard global average pooling (GAP) treats all spatial locations equally, mixing the discriminative eye-region signal with irrelevant activations. The EyeAttention module employs two consecutive 1×1 convolutions (1280 → 128 → 1) to obtain an attention logit map of size [B, 1, 7, 7]. A fixed anatomical prior tensor reflecting the expected vertical eye position within a normalised face crop (row weights: [0.5, 2.0, 3.0, 3.0, 2.0, 0.5, 0.3]) is element-wise multiplied with the learned logit map before softmax normalisation. The resulting weights produce an attended 1280-dimensional feature vector:

$$f\_vis = \Sigma(i,j)\ w(i,j) \cdot F(i,j), \quad w = softmax(conv(F) \times prior)$$

EAR Geometric Stream. In parallel, a 7-dimensional EAR feature vector is computed from MediaPipe FaceMesh landmarks using MediaPipe landmark indices for left eye [362, 385, 387, 263, 373, 380] and right eye [33, 160, 158, 133, 153, 144]. The standard EAR formula [7] is:

$$EAR = (\|P_2-P_6\| + \|P_3-P_5\|) \ / \ (2\|P_1-P_4\|)$$

The 7-dimensional feature vector comprises: left EAR, right EAR, mean EAR, |left−right| EAR, normalised left EAR, normalised right EAR, and a binary landmark detection flag. EAR values are normalised within [0,1] using EAR_closed = 0.20 and EAR_open = 0.28. When landmark detection fails a calibrated fallback vector [0.22, 0.22, 0.22, 0.0, 0.22, 0.22, 0.0] is substituted. The EAR vector is projected through two fully connected layers (7→32→64, ReLU) yielding a 64-dimensional geometric feature.

Fusion and Classification. The 1280-dim visual feature and 64-dim EAR feature are concatenated (1344-dim) and passed through a dropout-regularised classifier: Dropout(0.3) → Linear(1344, 256) → ReLU → Dropout(0.2) → Linear(256, 2) → Softmax, yielding P(Awake) and P(Sleeping).

### *E. Talking Detection: TalkingModel*

Talking detection is a binary classification (Talking / NotTalking) operating on the anatomically cropped mouth region. Using MediaPipe landmarks, the mouth bounding box is extracted, padded, and resized to 224×224; crops below 16×16 pixels are discarded. A pretrained MobileNetV2 backbone extracts [B, 1280, 7, 7] features, processed by the MouthAttention module (two 1×1 convolutions without anatomical prior, softmax-normalised over spatial positions). The attended 1280-dim feature is classified via Dropout(0.3) → Linear(1280, 256) → ReLU → Dropout(0.2) → Linear(256, 2) → Softmax, yielding P(NotTalking) and P(Talking).

### *F. Temporal Smoothing and Behaviour Hierarchy*

Frame-wise predictions are noisy because of transient expressions and micro-movements. A temporal smoother maintains rolling counters of the number of sleeping and talking predictions for the last few frames for each track id. A student is regarded as sleeping if the proportion of sleeping-prediction in the observation window is above a threshold. A strict hierarchy of behaviours is enforced. Sleeping is prioritized, so if a student is classified as sleeping, the talking model is not invoked, saving edge compute. Then the awake student is assessed for speech. This hierarchy ensures logical label consistency and eliminates contradictory outputs.

### *G. Training Strategy*

Training of SleepingModelV4 is performed in 2 phases. Phase 1 (5 epochs): freeze the backbone MobileNetV2, and train only EyeAttention, EAR projection and classifier with Adam at lr = 3⇥10 4 . Phase 2 (up to 25 epochs): fine-tune the whole network using cosine annealing and early stopping (patience=7). Class imbalance is handled with class-weighted cross-entropy loss and gradient clipping with max norm 1.0 to stabilize training. A custom SimulateGlasses augmentation (25% probability) generates overlays of glasses in frame-only, tinted and dark variants for better robustness to eye occlusion.

TalkingModel is trained with Adam with lr = $3\times10^{-4}$ and cross-entropy loss for 20 epochs. Augmentation consists of random resized cropping, horizontal flipping, rotation (±10°) and colour jitter. Both models are normalized with ImageNet (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225] ).

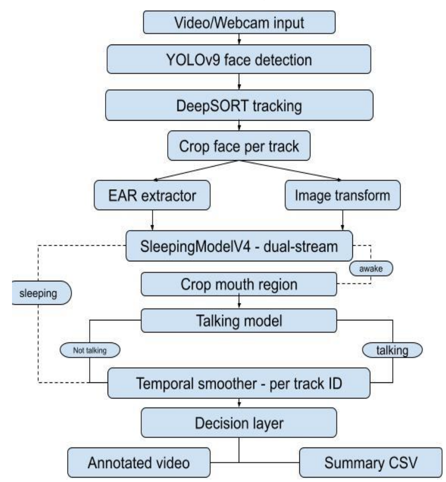


Fig 1: System Architecture

## IV. Dataset and Preprocessing

### *A. Sleeping Detection Dataset*

The sleeping detection dataset comprises face crops collected from publicly available sources, annotated into two classes: Awake and Sleeping. Images were standardised to 224×224 pixels. EAR features for each image were pre-extracted using MediaPipe FaceMesh and cached prior to training. For samples where landmark detection failed a calibrated fallback vector was applied. Training augmentation included random resized cropping (scale 0.7–1.0), horizontal flipping, rotation (±15°), brightness/contrast jitter (±0.4), random Gaussian blur (p = 0.2), random grayscale (p = 0.1), and SimulateGlasses (p = 0.25).

### *B. Talking Detection Dataset*

The talking detection dataset consists of mouth-region crops labelled as Talking or NotTalking, split into Train and Val subsets using PyTorch's ImageFolder format. Crops were extracted using MediaPipe outer-lip landmarks, padded by a fixed margin, and resized to 224×224.

### *C. Person-Level Test Videos*

End-to-end evaluation used five test video clips totalling 31.50 s at 25 fps (787 frames, 31 unique person tracks). Ground-truth behavioural labels were assigned per person per video segment by manual annotation. Two evaluation rounds were conducted: a single-video pilot (7 persons, 142 frames) and a five-video consolidated evaluation (31 persons, 787 frames).

## V. Experimental Setup

Model training was conducted on a GPU-accelerated environment (PyTorch 2.x, CUDA). SleepingModelV4: batch size 32, Phase-1 lr $3\times10^{-4}$, cosine annealing in Phase 2, early stopping patience 7, max 25 fine-tuning epochs, Adam optimiser, class-weighted cross-entropy loss. TalkingModel: batch size 32, lr $3\times10^{-4}$, 20 epochs, Adam, cross-entropy loss. Both models were initialised from pretrained ImageNet MobileNetV2 weights.

Deployment was validated on a Raspberry Pi 4 (4 GB RAM, CPU-only inference) running Python 3 with the ultralytics, deep-sort-realtime, mediapipe, opencv, picamera2, and flask libraries. Inference was configured at 1 Hz. MJPEG streaming was served via Flask at port 5000. Per-frame and per-person summary logs were exported as CSV.

## VI. Results and Discussion

### *A. Frame-Level Sleeping Detection*

Table I presents the frame-level results for SleepingModelV4 (397 validation samples). The model achieves 96.98% accuracy, 94.20% precision, and a perfect sleeping-class recall of 100% (zero false negatives). The AUC value of 0.9863 indicates good discriminability at all decision thresholds. The 12 false positives are students that are awake but their eyes look similar to the sleeping state due to partial occlusion, downward gaze or squinting.

**TABLE I. SLEEPINGMODELV4 FRAME-LEVEL PERFORMANCE**

| Metric | Value |
|---|---|
| Accuracy | 96.98% |
| Recall (Sleeping) | 100.00% |
| Precision (Sleeping) | 94.20% |
| F1-Score | 97.01% |
| Specificity | 94.06% |
| AUC | 0.9863 |
| False Negatives | 0 |

### *B. Frame-Level Talking Detection*

Table II presents TalkingModel results (583 validation samples). The model achieves 81.30% accuracy, 86.11% recall, 78.23% precision, F1 of 81.98%, and AUC of 0.8687. Lower performance relative to sleeping detection reflects the inherent ambiguity of talking behaviour: subtle lip movements from whispering, mouthing words, or yawning produce feature activations overlapping with the talking class, and speech intensity varies widely within the positive class.

**TABLE II. TALKINGMODEL FRAME-LEVEL PERFORMANCE**

| Metric | Value |
|---|---|
| Accuracy | 81.30% |
| Recall (Talking) | 86.11% |
| Precision (Talking) | 78.23% |
| F1-Score | 81.98% |
| Specificity | 76.61% |
| AUC | 0.8687 |

### *C. Person-Level Pipeline Evaluation*

Table III. Consolidated results at person level (31 persons, 787 frames, five videos) The pipeline achieved an overall accuracy of 77.42%, a weighted F1 of 0.7676, a precision of 0.8264 and a recall of 0.7549. Sleeping person-level recall is 100% (none of the sleeping students were assigned an awake or talking label at the person level for all test videos). The main source of errors is awake/talking confusion. The talking model sometimes classifies quiet talkers as awake, due to the minimal movement of their lips.

**TABLE III. PERSON-LEVEL PIPELINE RESULTS (5 VIDEOS, 31 PERSONS)**

| Metric | Value |
|---|---|
| Persons Evaluated | 31 |
| Total Frames | 787 |
| Video Duration | 31.50 s |
| Overall Accuracy | 77.42% |

| | |
|---|---|
| Weighted F1-Score | 0.7676 |
| Precision | 0.8264 |
| Recall | 0.7549 |
| Sleeping Recall (Person-Level) | 100% |
| ID Switches (DeepSORT) | 0 |


### *D. Comparison with Existing Methods*

Table IV compares the proposed system with prior works. The proposed approach uniquely combines multi-stream feature fusion (CNN + EAR), identity-preserving tracking-aware temporal smoothing, and validated Raspberry Pi deployment. No compared method reports 100% sleeping recall.

**TABLE IV. COMPARISON WITH EXISTING METHODS**

| Method | Task | Performance |
|---|---|---|
| Yang et al. [1] | Behaviour | mAP 49% |
| Zhang et al. [3] | Behaviour | mF1 53.18% |
| Zhang & Li [4] | Behaviour | Low in crowds |
| Gao & Hang [6] | Behaviour | ~95% (lab) |
| Ours (Sleep) | Sleeping | F1=97%, R=100% |
| Ours (Talk) | Talking | F1=82% |

### *E. Analysis of Modular Contributions*

The dual-stream design contributes measurably to sleeping detection robustness. The EAR stream provides lighting-invariant, pose-sensitive measurements that are informative when the CNN is uncertain — particularly when glasses partially occlude the eye region. The SimulateGlasses augmentation explicitly exposes training to such scenarios.

The EyeAttention anatomical prior ensures that, even when learned attention weights are uncertain, the effective receptive field is biased toward the upper-face rows of the feature map where eyes are anatomically expected. Without this prior, global average pooling would distribute equal weight across all 49 spatial positions including irrelevant mouth and forehead regions.

The temporal smoother's contribution is most visible at the person level. Frame-level accuracy of 96.98% would without smoothing produce occasional false labels integrated across an observation window. By accumulating evidence across frames with the behaviour hierarchy, the smoother elevates person-level sleeping recall to 100%.

### *F. Deployment Performance*

On Raspberry Pi 4 with CPU-only execution, the system operates at 1 Hz inference — sufficient for detecting sustained behavioural states that persist for seconds to minutes. The Flask MJPEG stream renders annotated frames continuously with bounding boxes, track IDs, labels, and probabilities overlaid. Per-frame and per-person CSV logs are available via the /export_csv endpoint.

## VII. Conclusion

This paper has presented a real-time, edge-deployable multi-student behaviour analysis system for emotionally intelligent robot teacher platforms. The pipeline integrates YOLOv9 face detection, DeepSORT identity-preserving tracking, the dual-stream SleepingModelV4 (MobileNetV2 + EyeAttention + EAR fusion), the MouthAttention-enhanced TalkingModel, and a behaviour-hierarchy temporal smoother. SleepingModelV4 achieves 96.98% frame-level accuracy with 100% recall and zero false negatives. The person-level pipeline achieves 77.42% accuracy across 31 students with zero identity switches. Deployment on Raspberry Pi 4 with a web-accessible real-time dashboard confirms practical edge deployability. These results establish that combining complementary visual and geometric modalities, spatial attention, identity-preserving tracking, and temporal smoothing produces a behaviour recognition system that is simultaneously more accurate and more practically deployable than prior approaches.

## VIII. Future Work

Future extensions include:

(1) integration of facial emotion recognition to classify valence and arousal states;

(2) audio-visual fusion using a co-located microphone to improve talking detection accuracy for whispered speech;

(3) transformer-based backbone replacements (e.g., EfficientFormer) for improved accuracy-to-latency trade-offs on ARM hardware;

(4) expansion of the behaviour taxonomy to include hand-raising, phone use, and note-taking; and

(5) closed-loop integration with adaptive learning management platforms enabling real-time adjustment of robot teaching strategies based on observed engagement patterns.

## Acknowledgements

The authors thank the Department of Electrical Engineering, Indian Institute of Technology Palakkad, and the OELP Coordinators for their continuous guidance and support throughout this research project.

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keywords: {Face recognition;Face detection;Noise;YOLO;Accuracy;Faces;Lighting;Image quality;Skin;Salt;Face detection and people counting;YOLOv9;crowd counting;object detection method;image denoising;speckle noise;salt and pepper noise;Gaussian noise;motion blur;filtering},